

Graph Coloring with $\Delta + 1$ Colors: Algorithms, Proofs, and Complexity

1 Introduction

An overview of a graph coloring algorithm for a graph with a maximum degree Δ can color using at most $\Delta + 1$ colors, assuming the EREW PRAM model.

2 Algorithms

2.1 Coloring Rooted Trees

The 6-Color-Rooted-Tree algorithm reduces the number of colors in a rooted tree iteratively:

1. Initialization:

- Assign each node in the tree a unique initial color equal to their IDs. It is essential for the IDs to be distinct numbers from 0 to $N - 1$, where N is the size of the graph.
- Each node knows the size of the graph, its ID, and Δ .

2. Color Refinement:

- Repeat until each node has at most 6 distinct colors:
 - Each node communicates with its parent.
 - The node compares its color with its parent's color bit by bit to find the smallest bit position where they differ.
 - It creates a new color by concatenating this bit position and the value of the differing bit.

3. Termination:

- When the number of colors is reduced to 6, stop the iterative process.
- The tree is now 6-colored.

The above can be refined further into a 3-Color-Rooted-Tree algorithm:

1. Reduce to Three Colors:

- Take the 6-colored tree and perform three sequential steps to reduce the number of colors to 3.
- In each step, recolor each node by checking the colors of its parent and children and choosing a new color that avoids conflict.

2.2 Coloring Constant-Degree Graphs

The Color-Constant-Degree-Graph algorithm extends tree-coloring to general graphs with degree Δ :

1. Partition into Pseudoforests:

- Identify a maximal pseudoforest within the graph.
- Remove the edges of the pseudoforest from the graph.
- Repeat this process until all edges are removed, partitioning the graph into Δ or fewer pseudoforests.

2. Color the Pseudoforests:

- Use the 3-Color-Rooted-Tree algorithm to color each pseudoforest in parallel.

3. Combine Colors:

- Reintroduce the edges of each pseudoforest one at a time.
- Recolor nodes to ensure consistency with adjacent nodes.

3 Theorems

Theorem 1: 6-Color-Rooted-Tree Algorithm Correctness

The 6-Color-Rooted-Tree algorithm produces a valid 6-coloring of a tree in $O(\log^* n)$ time on a CREW PRAM using a linear number of processors.

Proof:

- If node i after one step of reassigning colors had the same color as its parent, then that would mean they differed on the same i -th bit previously.
- Since the last bit of the new color is the value of the differing bit, they must still be different in at least one bit.
- The time complexity is $O(\log^* n)$.

Theorem 2: 3-Color-Rooted-Tree Algorithm Correctness

Proof:

- The number of colors is reduced to 3 by successive stages.
- Each stage shifts colors downward, eliminating one color.

- Each stage takes constant time.

Theorem 3: Coloring Constant-Degree Graphs

The Color-Constant-Degree-Graph algorithm colors a constant-degree graph with $\Delta + 1$ colors in $O((\log \Delta)(\Delta^2 + \log^* n))$ time on an EREW PRAM using $O(n)$ processors.

Proof:

- The graph is partitioned into Δ or fewer pseudoforests in $O(\Delta)$ time.
- Each pseudoforest is 3-colored in $O(\log^* n)$ time.
- Combining colors requires $O(\Delta^2)$ iterations.
- Total complexity: $O((\log \Delta)(\Delta^2 + \log^* n))$.

4 Complexity Analysis

- Rooted Tree Coloring: $O(\log^* n)$ time using $O(n)$ processors.
- Constant-Degree Graphs: $O((\log \Delta)(\Delta^2 + \log^* n))$ time using $O(n)$ processors.

5 References

- Goldberg, Plotkin, Shannon - Parallel symmetry-breaking in sparse graphs.